Tutorial 1

Question 1.4
A current of 7.4 A flows through a conductor. Calculate how much charge passes through any cross-sectional area of the conductor in 20 s.

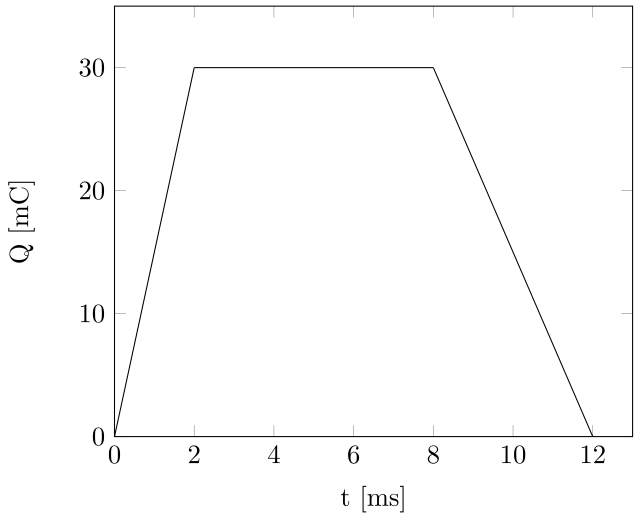
Question 1.6

The charge entering a certain element is shown below. Find:

a) $t = 1\ ms$

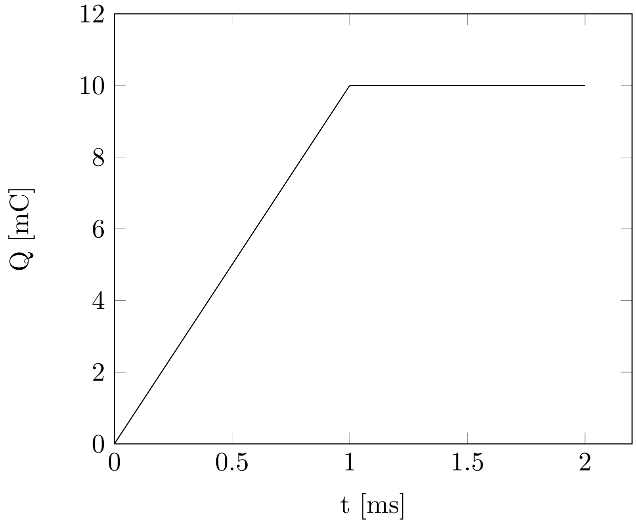
b) $t = 6\ ms$

c) $t = 10\ ms$



Question 1.8

The current flowing past a device is shown below. Calculate the total charge flowing past the point.



Question 1.11
A rechargeable flashlight is capable of delivering 90 mA for a period of about 12 hours. How much charge can it release at this rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

Question 1.12

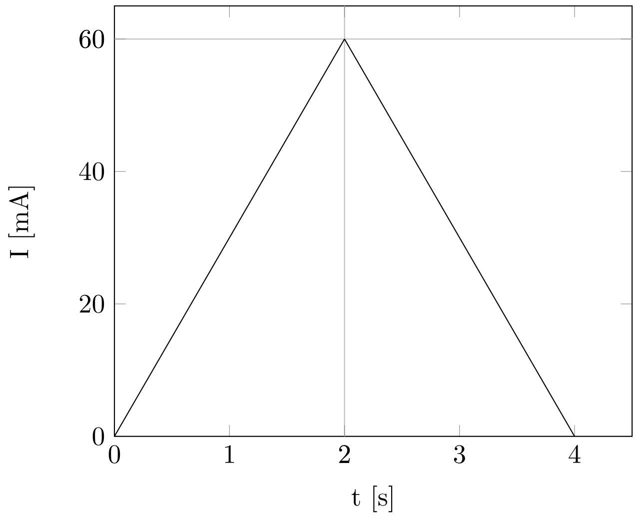
The current flowing through an element is given by

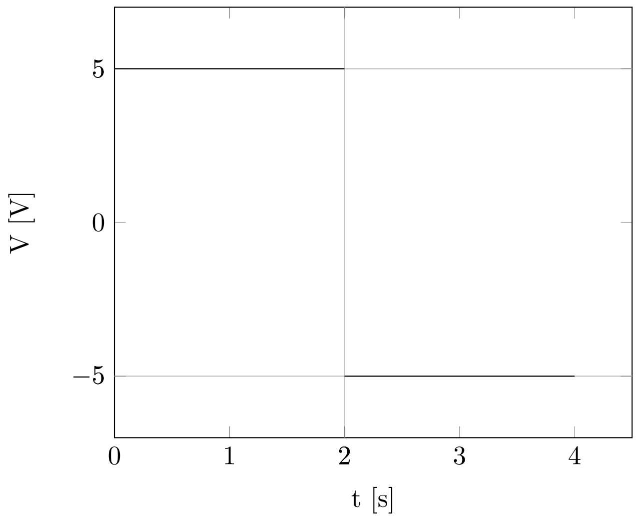
$$i(t) = \begin{cases} 3t\ A & 0 \leq t < 6\ s \\ 18\ A & 6 \leq t < 10\ s \\ -12\ A & 10 \leq t < 15\ s \\ 0 & t \geq 15\ s \end{cases}$$

Plot the charge stored in the element over $0 \leq t < 20\ s$.

Question 1.16

The following graphs show the voltage across and current through an element:



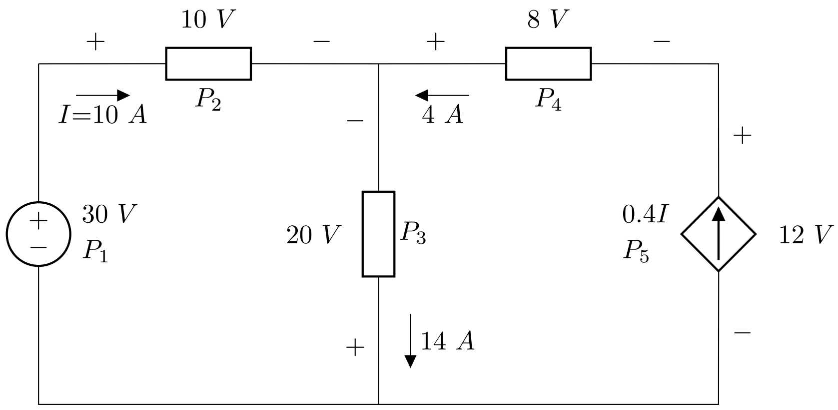


a) Sketch the power delivered to the elements for $t > 0$ s.

b) Find the total energy absorbed by the element for a period of $0 \leq t < 4$ s.

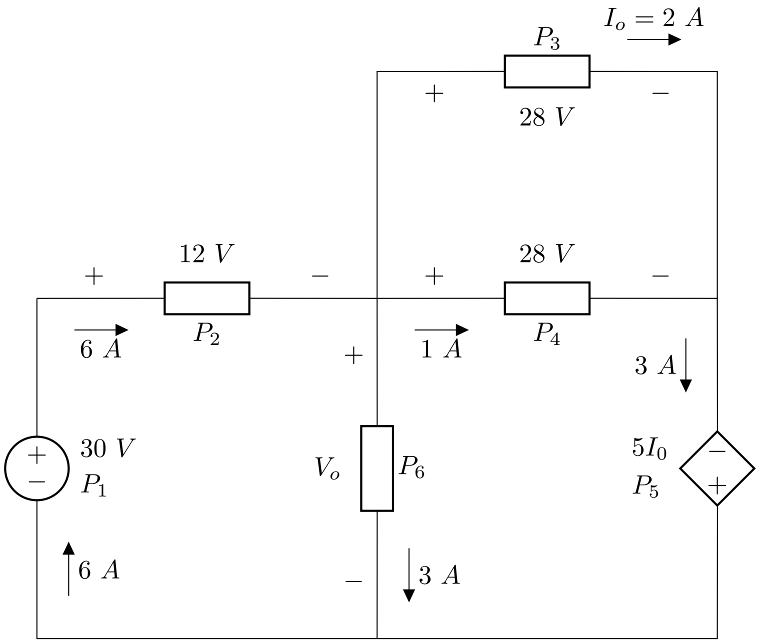
Question 1.18

Find the power absorbed by each of the elements.



Question 1.20

Find V_o and the power absorbed by each of the elements.



Question 1.24
A utility charges 8.2 cents/kWh. If a consumer operates a 60 W light bulb continuously for one day, how much is the consumer charged?

Question 1.28

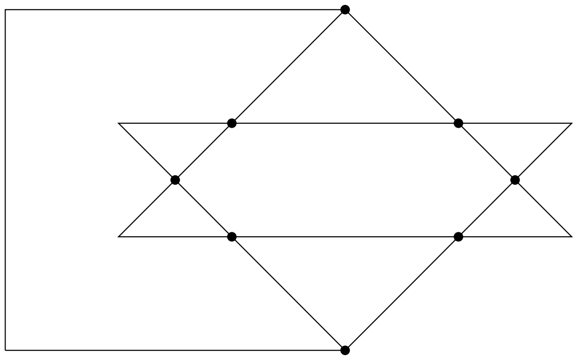
A 60 W incandescent lamp is connected to a 120 V source and is left burning continuously in an otherwise dark staircase. Determine:

a) the current through the lamp

b) the cost of operating the light for one non-leap year if electricity costs 9.5 cents per kWh.

Question 2.6

In the network graph below, determine the number of branches and nodes.



Question 2.7

Determine the number of branches and nodes in the circuit